[SB-Sg] Singapore Mapping, for Subway Builder

Tags	
DevTags	Data
Status	👩‍💻 Under Development

🇸🇬 Singapore Map Notes (reference)

TEMP

Todo

- first building/plot pass (initial data analysis) - visualize and tweak:
  - need to load both overture and MP2025
  - [ ] utility MP plot - what threshold to count as workplace
  - [ ] non-agri/utility MP plots that have less than [10%] of building footprint
    vary percentages and see what's more suitable
    - [ ] I would think something like 10% is better...
  - [ ] residential GPR plots is <2 a good value for res_med?
  - [ ] where B1/B2 - white

Links

EPSG:3414 - singapore CRS: https://spatialreference.org/ref/epsg/3414/

▼ Subway Builder Reference

SB Modding Docs

- https://www.subwaybuilder.com/docs/guides/custom-cities
- https://www.subwaybuilder.com/docs/api-reference/map#commercial--college-districts
- https://gist.github.com/ejfox/8ff61829030913aeb0e5fbcb1e1842e3 for `building_index.bin`
  - [ ] will need to do this at the very end
  - [ ] still need to check how it works, but that's something for later

SBM Registry Data Quality Docs

- https://subwaybuildermodded.com/registry/docs/data-quality/submission-questions

▼ Mapping Libraries

▼ Data Sources

Map - Divisions

- Regions, Planning Areas, Subzones
- "Singapore" is to use OSM geographic region

Map - Base Data

- **Buildings: Overture Foundation**
  - OSM + Various ML sources

    - OSM is already pretty good (and Overture takes priority from OSM) for Singapore, so Overture is really only used for data cleaning and
- **OSM** as a direct supplementary add-on -
    - [ ] primarily for `land_use` to verify against MP2025
    - not necessary to import buildings, those should be in the Overture map
- **Land Use: URA Master Plan 2025 Plots: interactive map | data**
    - See *MP25WrittenStatement.pdf* for land use categories
    - Known discrepancies (as this includes future zoning)
        - Special use - typically means military
        - Reserve site - typically means future development, but may have different current uses; Examples:
            - Keppel port areas
        - TJPG railway station (commercial with GPR=4.2)

▼ Other (unused)

- 2025 Building layer... not sure how useful, only govt buildings
- 2019 version: map | data

▼ OSM Notes

ambiguous tags - e.g. building:yes, or building:industrial

▼ MP Notes

List of zoning

- Residential
- Residential 1S Commercial
- Commercial & Residential
- Commercial
- Hotel
- White
- Business Park
- Business Park - White
- B1
- B2
- B1 - White
    - There may be minimum B plot ratios
- B2 - White
    - There may be minimum B plot ratios
- Residential/Institution
- Commercial/Institution
- Health & Medical Care
- Educational Institution
- Place of Worship
- Civic & Community Institution
- Open Space
- Park
- Beach Area

- Sports & Recreation
- Road
- Transport Facilities
- Rapid Transit
- Utility
- Cemetery
- Agriculture
- Port/Airport
- Reserve Site
- Special Use

Main Demand Modelling Workflow

▼ [etc] Utilities/Algorithms

how to find whether building resides in zone

- calculate by overlap?
- sometimes each plot is a building (esp HDBs), so might wanna do that

how to agglomerate?

Buildings (Residences, Workplaces)

▼ Pre-Processing

remove out of bounds buildings (rectangular bbox) - just in case pedra branca gets picked up

Removing buildings

- **Small**
  - default for depot is 40m^2
  - [ ] definitely watch this carefully (visualize)
  - [ ] histograms and maps
- **Unnecessary (tagged)**
  - ~~underground buildings: [OSM]~~ `location=underground` ~~,~~ `building:levels:underground=*` ~~, maybe some other tags on Overture~~ *include for demand modelling*
- **Unnecessary (zoning)**
  - remove all buildings in...
  - **OSM landuse**
    - construction
    - port → only use OSM buildings
  - **MP zoning**
    - Hotel - processed only in special demand
    - Health & Medical Care - major healthcare institutions included in special demand
    - Educational Institution - special demand
    - Place of Worship
    - Civic & Community Institution

        - Open Space
        - Park
        - Beach Area
        - Sports & Recreation
        - Transport Facilities
        - Rapid Transit
        - Utility, <50,000 sqm (arbitrary, should probably clear out everything but the large utility sites?)
            - Should probably check if it contains an (underground) reservoir - if it does, remove; I can probably hard code this. There are Fort Canning, Pearl's Hill, Mount Faber, Murnane
            - [ ] visualize and check
        - Cemetery
        - Port/Airport
        - Reserve Site
        - Special Use - special demand (miltary)
    - **Empty plots**
        - MP plots - remove plots with less than 10% building footprint, apart from agricultural and utility
            - [ ] to refine this number!

Assign buildings to MP Plots

- each remaining building gets a plot ID and plot zoning - easier processing

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▼ Extra: Manually-Processed Buildings

(or just added back)

Removed prior:

- Jewel
- SHOULDN'T BE REMOVED
    - PSA Horizons

Too complex to automatically handle:

- The Interlace

▼ Building Classification & Floorspace Estimation

> 💡 Produces new buildings at center points, with type+subtype and floorspace

Two main types: **residence/workplace**

- **residence**
    - res_low
    - res_med
    - res_high (HDB, EC, condo)
- **workplace**
    - office
    - commercial
    - ind_low

    - ind_med
    - ind_high

Building zone verification

- Checks whether a building actually belongs to its MP zoning, and assigns a corrected (if necessary) building type
  - [ ] TO REFINE
  - Reserve site & STDB (Subject to Detailed Planning)
  - go through every building:
    - if it roughly seems to match the MP zoning, nothing
    - else (how)?
      - remove building's connection to MP plot, set new zone
  - WIP WIP WIP WIP WIP WIP WIP WIP WIP
- If a building is assigned its MP zone
  - Residential
    - GPR=LND → res_low, else res_high
    - [ ] GPR<2 (or a different number, visualize and check) → res_med
  - Residential 1S Commercial
    - duplicate buildings, F=1 floor as commercial, F=remainder-1 floors as res_high
  - Commercial & Residential
    - duplicate buildings, 0.35*F as commercial, 0.65*F as res_high
  - Commercial
    - if any building tagged on Overture/OSM as office → office, else commercial
  - White
    - if any residentially tagged building on Overture/OSM → separate out as res
    - else → commercial
  - Business Park → office
    - Weird OSM tags:
      - `building=* office=*`
      - `building=* amenity=bank`
  - Business Park - White → office + commercial (if percentage stated, use that percentage, else 30%)
  - B1/B2
    - GPR=1 → ind_low
    - 2>GPR>1 → ind_med
    - GPR>2 → ind_high e.g. car dealerships (typically ~2.5)
  - B1 - White/B2 - White
    - [ ] might be worth checking if this even exists...
  - Residential/Institution → res_med
  - Commercial/Institution → commercial
  - Utility → ind_low
  - Agriculture → ind_low

Immediately after classification, Foorsplace estimation

- should be done in the same loop

- whether to take GPR or the actual building
- whether or not the building on Overture even has floor counts

- [ ] availability of floor/height data is TBC
- Residential: if foors are available, directly calculate floorspace
  - Buildings that lack floorcount: 0.9x GPR - calculated buildings
- Commercial/Office: if foors are available, directly calculate floorspace
  - Buildings that lack floorcount: 0.9x GPR - calculated buildings
- Upper bound by listed GPR

▼ Building Occupancy Estimation

To estimate a value that the population will be distributed into? or a max?

based on subtype and floorspace

▼ Floorspace examples

To check dwellings/sqm, go to https://eservice.ura.gov.sg/maps/?service=MP, explore development site, then look around clicking until you find one!

- Examples
  - The Terrace (EC): 81,820.32 sqm, 747 dwellings
  - Waterway sunbeam next door: 80,456.68 sqm, 661 dwellings
  - A random HDB estate in Toa Payoh (near the SAFRA): 351,439.73 sqm, 1218 dwellings... no that can't be right
  - Dakota Breeze: 113,228.15 sqm, 954 dwellings
  - The Interlace (might need this to be parsed manually manual): 180,663.47 sqm, 1040 dwellings
- Other GFAs
  - MBS + The Shoppes: 597,200.82 sqm

Known values & distributions

- These will be subtracted from remaining counts
- Residences - nil
- Workplaces
  - 138k Healthcare workers - see **special-demand;Healthcare**

Occupancy estimation

- multiply by factors

▼ Point Agglomeration

combine by distance

- [ ] probably have some algorithms, and especially something that is spatially efficient

Population Distribution

▼ Non-Resident Working Population Extrapolation

This is because datasets only count 15 y/o+ resident (Singapore citizens and permanent residents) employment

- it's missing another 1.91 million non-residents
- https://stats.mom.gov.sg/Pages/employment-Summary-Table.aspx
  - 4.117M total (including about 300k MDW), 2.481M resident, 1.319M non-resident
- https://www.mom.gov.sg/foreign-workforce-numbers:
  - 1.32 million workers (excl MDW) listed here
  - it says foreign workforce, so I presume this includes JB-SG commuters. And so that will have to be counted

Total Population Figures (figures from Population in Brief 2025, I can't find any more up-to-date data, the data.gov.sg MOM datasets cut out at 2022 and are *bad*)

Non-resident pass types percentages 2025

- **Employment Pass** ~11%
- **S-Pass** ~9%
- **Work Permit** ~44%
  - Non-CMP ~21%
  - CMP (*Construction, Marine (shipyard), Process*) ~23%
    - *if you're curious, NGO TWC2 estimate that 79% of that is construction*
- Migrant Domestic Workers - non-commuters, ignored
  - *Technically they're under WP as well*
- LTVP, Dependant's Pass - typically not employed, ignored
- Student pass - ignored, wrapped into education special demand

Handling Distribution

- It's not the easiest to differentiate between EP/SP and WP (non-CMP) working locations, so that 41% (783k) will just get distributed evenly
- WP (CMP) allows for more deliberate splitting:
  - likely will remove construction, assume 79% based on the TWC2 estimation, so that leaves 21% of WP (CMP) workers, about 92k

- [ ] cross border flows
  - Cross border flows are another residence source that have to be accounted for.
  - [ ] are these numbers included above...?
    - I still am not sure, but I think the answer is **no**
    - [ ] so I will have to subtract it
  - [ ] There are two options: ignore, or just place a mega resident source point on the checkpoints. The latter is quite funny, but I'm not sure if I want to do that.
  - Numbers
    - Total: 450,000 (Wikipedia)
    - via Woodlands: 300,000 [source] to 350,000 [original source not found]
    - via Tuas: I'm not sure, but if the Wikipedia top 10 list is correct, less than 100,000 daily

▼ Commute Flows

Gravity models? IPF? (Furness/IPF, exponential distance decay)?

▼ OD Constraints

Two main possible:

- Train/bus O/D demand - it's accurate, but might not be that useful - problems:
  - very hard to combine train and bus; compromise could be to use them separately
  - problems with very dense (CBD), or sparse areas with no MRT connection (Tuas, South Jurong)
- Easier direct option
  - Residential planning region → workplace planning area dataset, just use this

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Special Demand

▼ Healthcare

> 💡 This seeks to model both employment and hospital visit demand
>
> - Employment is reinserted at the building occupancy estimation level
> - Hospital visit demand is added demand

Key initial stats:

- https://en.wikipedia.org/wiki/List_of_hospitals_in_Singapore - compilation, I updated some of the figures & citations, but most are from 2021
- https://www.moh.gov.sg/others/resources-and-statistics/health-manpower/ up till 2024, broken down into many categories
- https://www.moh.gov.sg/healthcare-at-a-glance/# (2025) - workforce (138,000), public beds (14,333)

  > *workforce numbers - total registered healthcare professionals + non-registered works in PUBLIC institutions*

- Admissions
  - https://www.moh.gov.sg/others/resources-and-statistics/admissions-and-outpatient-attendances/
    - broken down into: **Admissions** (inpatient - for acute, psychiatric, community hospitals), Specialist outpatients, A&E, Polyclinics, Dental clinics, Day surgeries
    - and public/non-public
  - ^ as a singstat table, updated to 2025
  - BOR - not that useful

Major Healthcare Organisations

- [PUBLIC] SingHealth
  - Very detailed data in **annual report**
  - **SGH Outram** Campus:
    - SGH itself
    - NCCS, NDCS, NHCS, NNI, SNEC
    - [ ] likely also include the ~500 students from Duke-NUS... or as a separate point???
  - Acute hospitals: KKH, CGH, SKH
  - Communtiy hospitals: OCH (Outram - at SGH), SCH (Sengkang - At SKH)
- [PUBLIC] National Healthcare Group
  - **Annual report** has total statistics
    - Outpatient: **TTSH**, **NCID**, KTPH, IMH, WH, **NSC**, **AdMC**

        - Inpatient: **TTSH**, **NCID**, KTPH, IMH, WH, WH-CH, **ICH**, YCH
    - Acute hospitals: TTSH (includes NCID, NSC, ICH), KTPH, WH (Woodlands Hospital)
    - Community hospitals: WH-CH, YCH (Yishun Community Hospital)
    - Specialty centres: NCID (at TTSH), NSC (at TTSH), IMH
    - Extra: AdMC (Admiralty Medical Centre)
- [PUBLIC] National University Healthcare System
    - per-hospital beds, NUH has approximate staff
    - AH, NUH, NTFGH
- [PRIVATE] IHH Healthcare (prev known as Parkway Pantai)
    - https://www.ihhhealthcare.com/investors/financial-information/operational-highlights 932 licensed beds; 56,385 inpatient admissions
    - https://www.ihhhealthcare.com/sg: 1000+ beds & 5400 staff
- [PRIVATE]
    - Crawfurd Hospital, Farrer Park Hospital, Raffles Hospital, Thomson Medical Centre
- [NON-PROFIT]
    - Mount Alvernia - counted as **acute**
    - **Community**:
        - Thye Hua Kwan Hospital, St. Andrew's Community Hospital - Wikipedia
        - St. Luke's Hospital - **Annual report** has all figures required :) ... except for beds
        - Ren Ci Community Hospital - no official bed figure. Their website says 1301 beds across the hospital and its 3 nursing homes. Nevermind, by subtracting the nursing homes, I get 243 beds
- Other Hospitals
    - IMH 1950 beds (website), no staffing figures

Counting patient visit demand

- outpatient + inpatient?
- AE? Prolly not
- ignore day surgeries

☐ likely have to crunch the numbers then only see how the values look

- Extrapolation - see Yukina's maps as examples:
    - Sapporo (JP Example): OUT = 0.75*BEDS; DEMAND = 1.5*BEDS
    - Taipei (TW Example): OUT = 3.6*BEDS; DEMAND = 4.3*BEDS
    - Looks like they ***probably*** distributed some high level value amongst healthcare institutions, that would explain the large differences; or there's more precise decision making going on

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▼ Education

💡 ☐ maybe also polytechnics, cuz they large, and probably have official enrollment

and ignore staff

▼ Universities

> only public universities

Enrolment by Institutions (latest 2024) | data.gov.sg - helps when data is incomplete

NUS (31086 UG + 21765 GD) - might not include all GD schools

- Separate UG and GD stats
- Faculties:
  - CDE
  - SoC
    - ComEng split b/w CDE and SoC
  - FoS
  - FASS
  - Business
  - Law (now at UTown)
  - YSTCM
  - Dentistry + YLLSM + Saw Swee Hock School of Public Health (postgrad only)
  - LKY (postgrad only), still at **Bukit Timah**
    - 386 students...
  - Duke-NUS (postgrad only), at **SGH Outram**
    - 500 students - maybe combine with the SGH Outram campus healthcare point
  - Likely combined into the following points:
    - CDE + YSTCM
    - FASS + Biz + SoC
    - FoS + MD + etc.
    - UTown (Law, *ideally also NUSC but I don't think those numbers are a thing*)
- Housing: student housing site quotes 11000: https://uci.nus.edu.sg/campus-life/housing-services/student-housing/
  - then distribute based on OSM/Overture building data (it looks pretty complete to me)
  - ^ must allow housing points to agglomerate with nearby standard residence points as well

NTU (25010 UG + 15729 GD) - might not include all GD schools

- https://www.ntu.edu.sg/about-us/facts-figures/enrolment-by-college
  - Engineering, Science, Nanyang Business School, Humanities Arts & Social Sciences, Med, Comp/DataScience, NIE
  - trying to geographically pin this may be more of a challenge
- no figures on housing, likely have to estimate based on NUS' coefficients
- https://maps.ntu.edu.sg/

SMU (2022 stats, guh)

- 9883 UG, 3166 PG
- The school actually has quite a large campus (see map), so if you can find per-faculty statistics, that would be nice
  - or not

- 407 faculty :)

SUSS

- Map
- MOE 2024 stats: 4154 + 2570????
- website says 19000+, ST says 30000, though with a third being CET courses

SIT

- MOE 2024 stats: 10811 + 4417

SUTD

- MOE 2024 stats: 1969 + 699

UAS

- MOE 2024 stats: NIL

▼ Polytechnics

> 💡 I think these should suitably be counted as college campuses

Use Enrolment by Institutions (latest 2024) | data.gov.sg

Also Lasalle, NAFA, ITE

▼ Tourism

Hotels

- note that the OSM tag is `building=* tourism=hotel`

Attractions

Demand Linking

Extra Notes

Geographical Distribution - PowerBI

[2025 GHS Report] Western Water Catchment - 54.6% <35y employed, ~70% there were military personnel

Buildings

Map Geometry

> 💡
> - [ ] just build with depot????
>   - at the bery least, looking at those configs helps
>   - especially potential optimization options
>   - I mean, honestly, maybe just do this for ease, at least a first publish

if you want to use depot and exclude train station collisions, that option is currently pending a PR

▼ Visual Footprint Parse

> 💡 For footprint collisions

Again, by building size

then basically just remove construction but keep everything else

and transportation? railway??? stations, specifically?

Additional Layers

☐ coastline and water rendering seems to be something that generally needs fixing, looking at yukina's maps' changelogs

▼ Foundations

- Use building parts to generate if possible
  - what comes to mind are the CCL tunnels slightly clipping the outer sections of the MBS footprint
  - `building:has_parts` may be useful for deciding - I've seen some buildings in the Singapore downtown use that
- may still not be ideal because a lot of major buildings are built with circular caisson foundations

clamping to -10, -80

see the various MRT construction papers for examples

☐ possible... cutting foundation depths for known MRT routes...?

▼ Bathymetry

GEBCO (15 arcsec, ~400m) is really only the only public (that I can legally use, for free) dataset for this

Other options:

- MPA, printed charts or ENC: https://www.mpa.gov.sg/port-marine-ops/marine-services/charts-tidal-info-atons-and-hydrography
- https://www.sciencedirect.com/science/article/pii/S0141118725001610 used C-Map https://www.c-map.com/chartexplorer/ charts, they even have the proper high-resolution charts publicly viewable, but extracting that and trying to read it wil be hell, not to mention it's probably outside ToC :)
- a different paper (https://www.researchgate.net/publication/389975862_Forced_and_free_waves_of_simulated_volcanic_meteo-tsunamis_in_the_South_China_Sea#pf9) used https://www.garmin.com/en-US/p/1982453/pn/010-C1493-40/ - USD 400

> The Singapore high-resolution bathymetry grid derived from a digitation exercise of water depth data points from local nautical charts[26], combined with direct water depth data points from GEBCO 2023, land elevation points from the 30 m SRTM dataset[24] and 10 m DTM dataset of Singapore, all corrected to mean sea level using local tide gauges data, to obtain a final 50 m resolution grid.

as for reservoirs, the large Singapore reservoirs have their capacity on Wikipedia, you could use that to calculate average depths or smth :)

▼ OD Driving Paths

☐ driving paths

Likely something like OSMnx

There is a Onemap API (https://www.onemap.gov.sg/apidocs/routing/#walkdrivecycle), but that surely won't work for this

▼ Not Included

Departure times - these are estimated by an in-SB algorithm, based on a travel demand schedule

Final Matters

Licencing, attribution: https://data.gov.sg/open-data-licence

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Future Notes

- Additional reference/ideas: https://buildings.sg/

Singapore data

https://nusgis.org/data/

NUS UAL: https://ual.sg/project/open-urban-data-singapore/

Rest of SEA: https://ual.sg/post/2020/09/12/new-paper-exploration-of-open-data-in-southeast-asia-to-generate-3d-building-models/